

Name:	Class:
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Grade Eight Science Criterion D Assessment - Light and Optics PRACTICE Case Study 1

INSTRUCTIONS:

- Read the Practice Case Study 1 below taken from the [website Physics Case Studies](https://www.physicscasestudies.org/periscope/) from the following url: [https: www.physicscasestudies.org/periscope/](https://www.physicscasestudies.org/periscope/)
- Answer the questions associated with the case study
- See the last page for the command terms and Criterion D rubric if needed

UNKNOWN WORD BANK:

Periscope: • An optical instrument used to observe objects from a concealed position	Optronic Mast: • A high tech digital periscope
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OPTICS PRACTICE CASE STUDY 1: THE SILENT EYE - ENHANCING SUBMARINE NAVIGATION
By. Andrew MacDonald

In the early 20th century, naval engineering faced a massive hurdle. Submarines allowed navies to remain hidden underwater, but once submerged, the crew was effectively "blind." To see the surface, a submarine had to rise dangerously close to the waves, risking detection by enemy ships.

The solution was the **periscope**. A standard periscope uses a long tube with two **plane mirrors** positioned at 45-degree angles at each end. Light from an object on the surface hits the top mirror, reflects downward at a 90-degree angle, and then hits the bottom mirror, which reflects it again into the observer's eye.

Traditional glass mirrors served sailors well for decades, but they had limitations. In rough seas, vibrations would often cause the mirrors to shift, distorting the image. Furthermore, glass mirrors lose about 5-10% of light through absorption, making them difficult to use at dawn or dusk. A high-grade naval glass periscope costs approximately \$15,000 to manufacture.

Recently, engineers have moved toward optronic masts. These replace physical mirrors and tubes with high-definition digital cameras and fiber optics. While this allows for "night vision" and thermal imaging, it makes the submarine vulnerable to cyber-attacks and electronic failure. If the power goes out, the digital periscope is useless, whereas a manual mirror periscope requires no electricity to function. Also, a single optronic mast costs roughly \$450,000—thirty times the price of a mirror system.

Although optronic mast technology is more expensive, an issue with traditional periscopes is that the production of high-quality silvered mirrors involves the use of chemicals like silver nitrate and copper. If manufacturing waste is not managed correctly, these heavy metals can leach into local water systems, harming aquatic life. If a factory leaks 1kg of silver nitrate into a local stream, it can kill 90% of the local fish population within 48 hours and can cost \$120,000 per kilometer to clean up. However, the use of periscopes has also been adapted by marine biologists to study coral reefs without disturbing the delicate ecosystems with large diving equipment.

MLA Citation:

- Use the MLA quick reference guide to cite the Case Study in the box below

QUESTIONS - CASE STUDY 1 (PRACTICE):

1. **Describe** how the development of the optronic mast (a scientific advancement) was intended to solve an environmental problem associated with traditional periscopes

The development of [scientific advancement] was intended to address [environmental problem]. This problem is caused by [scientific cause], which results in [scientific change].

According to the case study, “[direct quote]” (Author).

[scientific advancement] works by [mechanism], where [scientific explanation]. This leads to [direct effect]. As a result, [final outcome], which helps to [solve the problem].

2. **Analyse** any two scientific implications of using "silvered mirror," technology

Implication 1	Implication 2
One significant scientific implication is the absorption and distortion of light. Because traditional glass mirrors lose 5-10% of light through absorption, their functional utility is limited in low-light conditions (dawn or dusk). Furthermore, because the system relies on the law of reflection—where the angle of incidence must equal the angle of reflection —any mechanical vibration in rough seas that shifts the mirrors at the 45-degree points will cause image distortion. This makes the physical alignment of the mirrors a constant technical vulnerability.	A positive implication of this technology is its passive nature. Unlike optronic masts, which require a consistent power supply to function, silvered mirrors rely only on the physical properties of light reflection. In a "power-out" scenario or during an electronic failure, the mirror system remains fully functional. This creates a high level of mechanical reliability; while it lacks "night vision" capabilities, it is immune to the modern threats of cyber-attacks or electrical circuit failure, ensuring the crew is never truly "blind" as long as there is ambient light.

3. **Discuss** one positive and one negative implication of using "silvered mirror" technology considering the social factor

One Positive	One Negative
Accessibility and Education: The relative affordability and simplicity of mirror-based periscopes have a positive social impact on the scientific community and the public's understanding of nature. Because these systems are thirty times cheaper than optronic masts, they have been successfully adapted for marine biology. This allows researchers and educators to study delicate coral reef ecosystems without the intrusive presence of heavy diving gear. Socially, it gives more people access to marine observation, fostering a more "ocean-literate" society and encouraging the conservation of biodiversity through non-invasive observation.	Public Health and Safety: The manufacturing process for silvered mirrors carries a severe risk to local communities. If a production facility fails to contain heavy metals like silver nitrate, the resulting contamination of local waterways (\$120,000 per kilometer to clean) directly threatens the food security and health of the surrounding population. Socially, a single industrial accident could destroy the livelihoods of local fishing communities and lead to long-term health crises due to heavy metal bioaccumulation in the water supply. In this context, the "low cost" of the \$15,000 periscope is offset by a massive potential social cost to the public.

MYP Command Terms that are used on Criterion D:

Summarize: Abstract a general theme or major points	Outline: Give a brief account or summary
Apply: Use knowledge and understanding in response to a given situation	Describe: Give a detailed account or picture of a situation, event, pattern, or process
Discuss: Offer a considered and balanced review that includes a range of arguments, factors, or hypotheses. Opinions or conclusions should be presented clearly and supported by appropriate evidence	Explain: Give a detailed account including reasons and causes
Evaluate: Make an appraisal by weighing up the strengths and limitations	Identify: Provide an answer from a number of possibilities.

Rubric: Criterion D: Reflecting on the Impacts of Science

Achievement Level	Level Descriptor
0	The student does not reach a standard identified by any of the descriptors below
1-2	The student is able to: state the ways in which science is applied and used to address a specific problem or issue state the implications of using science and its application to solve a specific problem or issue, interacting with a factor: social, environmental, economic, cultural, or ethical apply scientific language to communicate understanding but does so with limited success - documents sources with limited success
3-4	The student is able to: outline the ways in which science is applied and used to address a specific problem or issue outline the implications of using science and its application to solve a specific problem or issue, interacting with a factor: social, environmental, economic, cultural, or ethical sometimes apply scientific language to communicate understanding sometimes document sources completely
5-6	The student is able to: summarize the ways in which science is applied and used to address a specific problem or issue describe the implications of using science and its application to solve a specific problem or issue, interacting with a factor:social, environmental, economic, cultural, or ethical usually apply scientific language to communicate understanding clearly and precisely usually document sources completely
7-8	The student is able to: describe the ways in which science is applied and used to address a specific problem or issue discuss and analyse the implications of using science and its application to solve a specific problem or issue, interacting with a factor: social, environmental, economic, cultural, or ethical consistently apply scientific language to communicate understanding clearly and precisely - document sources completely

PART A. CITING IN-TEXT CITATIONS

1. **Citing with a known author:** (Lopez, 87) or (Lopez)
 - Use the last name of the author and page number
 - Use the last name of the author only if you don't have a page number)
2. **Citing with an unknown author:** ("Green Energy Use")
 - Use the name of the article in quotes inside a bracket

PART B. CITING FOR YOUR MLA WORKS CITED

1. Website with One Author

Format:

Last name, first name. Title of the article in quotation marks. The name of the website in italics. Date of Use. Web Link.

Sample:

Gadbery, Elizabeth. "The Best Houseplants for Low Light." *The Spruce*, 14 Jan. 2026, www.thespruce.com/best-low-light-houseplants-4774244.

2. Website with More Than One Author

Format:

Last name, first name and First name, Last name. Title of the article in quotation marks. The name of the website in italics. Date of Use. Web Link.

Sample (Two Authors):

Miller, Julian, and Sarah Chen. "Climate Trends in the Pacific Northwest." *Environmental Watch*, 2 Feb. 2026, www.envirowatch.org/climate-trends-pnw.

3. Website with More Than Two Authors

Format:

Last name, first name. Title of the Website in quotation marks. The name of the website in italics. Date of Use. Web Link.

Gadbery, Elizabeth at al. "The Best Houseplants for Low Light." *The Spruce*, 14 Jan. 2026, www.thespruce.com/best-low-light-houseplants-4774244.

4. Website with No Author

Format:

Title of the Website in quotation marks. The name of the website in italics. Date of Use. Web Link.

Sample:

"History of the Golden Gate Bridge." *History.com*, A&E Television Networks, 15 May 2025, www.history.com/topics/landmarks/golden-gate-bridge.

5. YouTube Video or Video from Other Source

Format:

Creator's Name or Channel Name. Title of the Video in Italics. The name of the website in italics. Date of Use. Web Link.

Sample:

Physics Girl. "Why the Sky is Blue (And Why it's Sometimes Green)." *YouTube*, 10 Nov. 2024, www.youtube.com/watch?v=bcdefg12345.

6. AI-Created Work

Format:

Prompt in quotation marks followed by the word prompt. AI Tool. Date of Use. Web Link.

Sample:

"Explain the concept of quantum entanglement in simple terms" prompt. *Gemini*, 3 Flash version, Google, 11 Feb. 2026, gemini.google.com.